Checkpoint 6 Maths — Practice Test 3

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 40 marks

1 At the school fun fair there is a "guess the mass" game. The cake weighs 3.586 kg. **[1]**

What is the value of the digit 8 in 3.586?

- ☐ A) 8 ones
- ☐ B) 8 tenths
- ☐ C) 8 hundredths
- ☐ D) 8 thousandths

2 A farm keeps its milk and meat in cold rooms. **[1]**

(a) The meat room is at −7 °C. The temperature rises by 12 degrees. What is the new temperature?

Working

Answer _____ °C

(b) The milk room is at 4 °C. The freezer is at −15 °C. What is the difference between the two temperatures? **[1]**

Working

Answer _____ degrees

3 Karim counts back in equal steps. **[1]**

1.4, 0.9, 0.4, _____, −0.6

Write the missing number.

Working

Answer

4 A bookshop orders 24 boxes of bookmarks. Each box has 1 367 bookmarks. **[2]**

How many bookmarks does the shop order altogether? Show your working.

Working

Answer

5 A farmer collects 756 eggs. She packs them in trays. Each tray holds 18 eggs. **[2]**

How many trays does she fill? Show your working.

Working

Answer trays

6 **[1]**

(a) Which calculation has the answer 29?

- ☐ A) 4 + 6 × 3 − 1
- ☐ B) (4 + 6) × 3 − 1
- ☐ C) 4 + 6 × (3 − 1)
- ☐ D) (4 + 6) × (3 − 1)

(b) Work out $50 - 8 \times 5$ **[1]**

Working

Answer

---

7 Rana makes prize bags for the fun fair. She has 24 red balloons and 36 blue balloons. **[1]**
Every bag must have the same number of red balloons and the same number of blue balloons. No balloons are left over.

What is the greatest number of bags she can make?

Working

Answer bags

---

8 **[1]**

(a) Which number is divisible by 9?

- ☐ A) 2 571
- ☐ B) 5 162
- ☐ C) 7 235
- ☐ D) 3 474

(b) Tarek says, "Every number that is divisible by 3 is also divisible by 6." **[1]**
Give an example to show that Tarek is wrong.

---

9 **[1]**

Write the cube number that is between 100 and 130.

Working

Answer

10 Amir builds model cars in a workshop. Each car needs 4 wheels. He always orders 3 spare wheels as well. **[1]**

The rule for his order is:

number of wheels = 4 × number of cars + 3

(a) How many wheels does he order for 9 cars?

Working

Answer ☐ wheels

(b) Mona says, "Amir ordered exactly 50 wheels." **[1]**

Explain why Mona must be wrong.

11 A model car is 100 times shorter than the real car. The real car is 4.2 m long. **[1]**

How long is the model car, in metres?

Working

Answer ☐ m

12 An axle for a model car is 6.24 cm long. **[1]**

Complete: 6.24 = 6 + 0.1 + _____

Working

Answer ☐

13 A farmer shares 7 bales of hay equally between 4 horses. **[1]**

How many bales does each horse get? Give your answer as a fraction or a mixed number.

Working

Answer _____ bales

14 On the farm, 18 of the 24 cows are black and white. **[1]**

What fraction of the cows are black and white? Give your answer in its simplest form.

15 A farmer plants 1/3 of a field with wheat and 1/4 of the field with maize. The rest of the field is grass. **[2]**

What fraction of the field is grass? Show your working.

Working

Answer _____

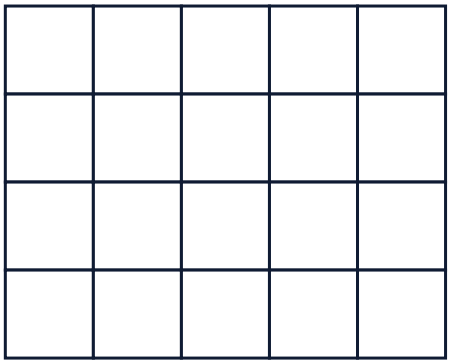
16 A bag of rabbit food holds 3/4 kg. It is shared equally between 3 rabbits. **[1]**

How much food does each rabbit get, in kilograms?

Working

Answer _____ kg

17 A bookshop is making a sale poster on this grid. **[1]**



Shade 35% of the grid. (Copy the grid on paper.)

18 **[1]**

(a) Write <, > or = to make this correct:
3/4 ☐ 0.8

(b) Sara says, "0.45 is less than 2/5." **[1]**
Is Sara correct? Explain how you know.

19 At the swimming gala, Laila swims 50 m in 31.25 seconds. Nour swims 50 m in 29.8 seconds. **[1]**

How many seconds faster is Nour than Laila?

Working

Answer ☐ seconds

20 A notebook in the bookshop costs $2.35. **[1]**

How much do 6 notebooks cost?

Working

Answer $

21 The swimming gala lasts 3.25 hours. **[1]**

Complete: 3.25 hours = 3 hours and _____ minutes

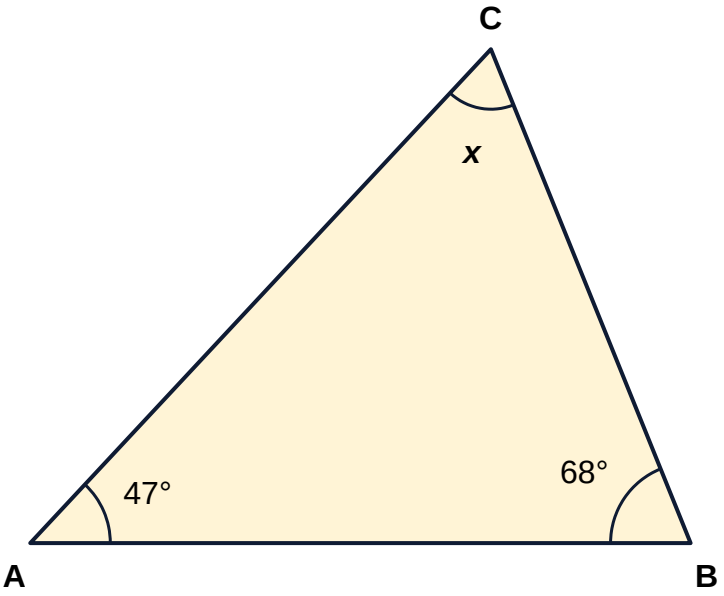
Working

Answer minutes

22 Amir wants a paper wheel for a model car. **[1]**

Use compasses to draw a circle with a **diameter** of 7 cm. Mark the centre and draw one radius. Label it "radius".

23 A triangular flag hangs at the fun fair. The diagram is not drawn to scale. **[1]**

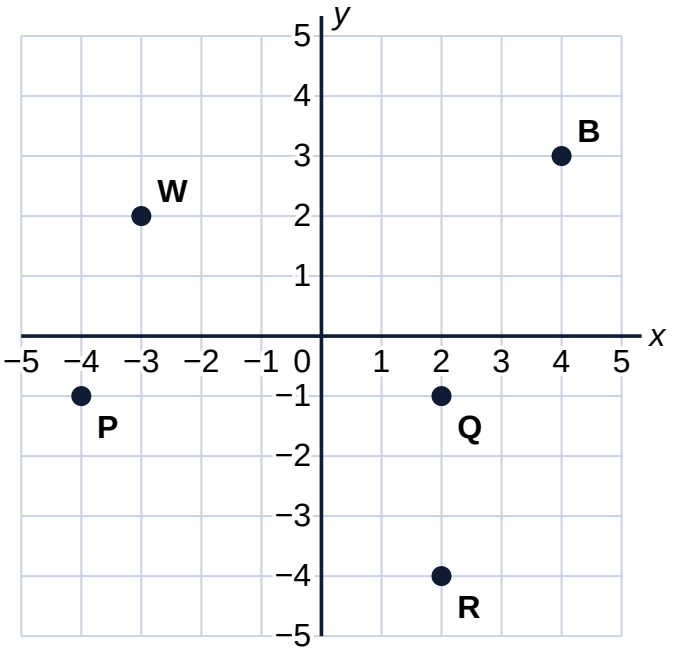


Work out the size of angle x.

Working

Answer ____ °

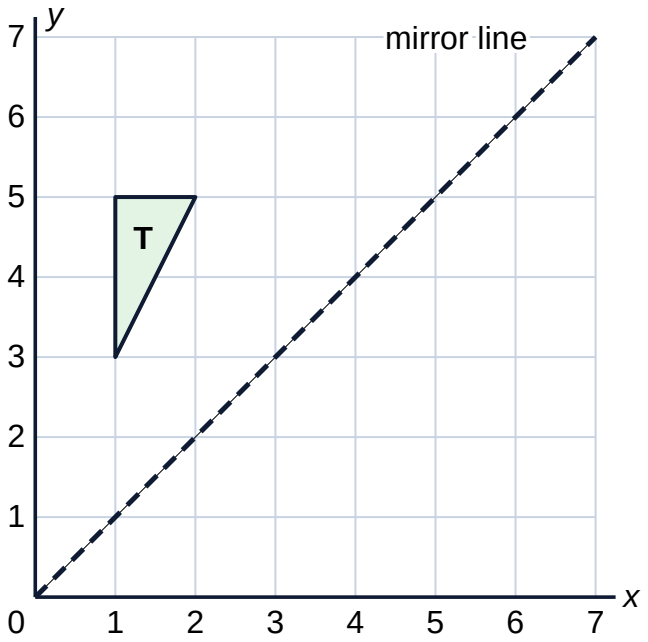
24 This map of a farm is drawn on a coordinate grid. W is the well and B is the barn. **[1]**



(a) Write the coordinates of the well, W.

(b) P, Q and R are three corners of a rectangular field. Write the coordinates of the fourth corner. **[1]**

25 The diagram shows a triangle T and a mirror line. **[1]**

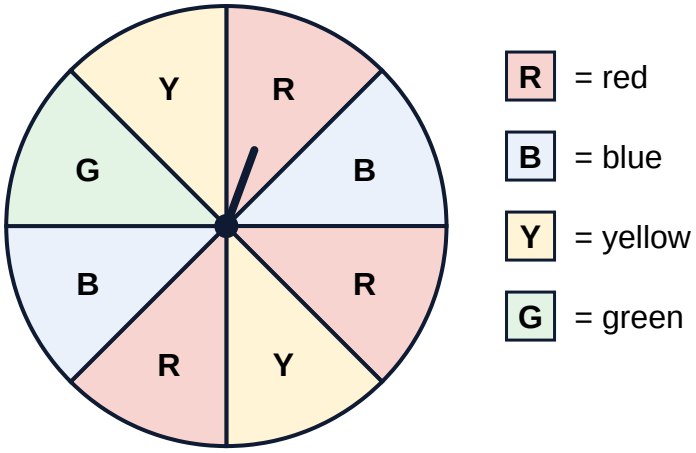


Draw the reflection of triangle T in the mirror line. Write the coordinates of the corners of your reflection. (Copy the grid on paper.)

26 **[1]**

Huda says, "A square is a special kind of rhombus."
Is Huda correct? Explain your answer.

27 This spinner is used for a game of chance at the fun fair. All 8 parts are the same size. **[1]**



(a) Which word best describes the chance of the spinner landing on red?

- ☐ A) unlikely
- ☐ B) even chance
- ☐ C) likely
- ☐ D) certain

(b) Can "landing on red" and "landing on blue" happen at the same time on one spin? Write the name for events like these. **[1]**

28 Five swimmers in Team A swim one length. Their times are: **[1]**

Swimmer	Ali	Omar	Hany	Ziad	Adam
Time (seconds)	41	38	45	39	43

(a) Find the median time.

Working

Answer ____ seconds

(b) The range of Team A's times is 7 seconds. Team B's times have a range of 12 seconds. Which team's times are more consistent (closer together)? Give a reason. **[1]**

29 The farm records how many eggs are collected on six days: **[1]**
12, 15, 18, 27, 30, 36

Copy and complete the Carroll diagram by writing each number in the correct box.

	Multiple of 9	**Not a multiple of 9**
Even		
Not even		

Paper 2 — 45 minutes — 40 marks

1 At the swimming gala, Yasmin's time is 34.76 seconds. **[1]**

Round 34.76 to the nearest tenth of a second.

Working

Answer ____ seconds

2 A sign in a bookshop says: "3/10 off all atlases". **[1]**

Which of these is the same as 3/10?

- ☐ A) 3%
- ☐ B) 0.03
- ☐ C) 30%
- ☐ D) 0.13

3 In the bookshop sale, a dictionary normally costs $24. The sale price is 30% less. **[2]**

Work out the sale price of the dictionary. Show your working.

Working

Answer $____

4 Last month the bookshop sold 36 atlases. This month it sold 5/4 of that number. **[1]**

How many atlases did it sell this month?

Working

Answer ____ atlases

5 The bookshop's machine prints 4 896 bookmarks each day for 31 days. **[1]**

Which is the best estimate of the total number of bookmarks printed?

- ☐ A) 150 000
- ☐ B) 15 000
- ☐ C) 1 500 000
- ☐ D) 1 500

---

6 In the model-car workshop, 3 model cars need 45 ml of paint. **[1]**

(a) How much paint is needed for 7 model cars?

Working

Answer ____ ml

(b) A tin holds 250 ml of paint. How many model cars can be painted completely with one tin? **[1]**

Working

Answer ____ cars

---

7 On a farm, for every 2 cows there are 7 chickens. **[1]**

(a) There are 63 chickens. How many cows are there?

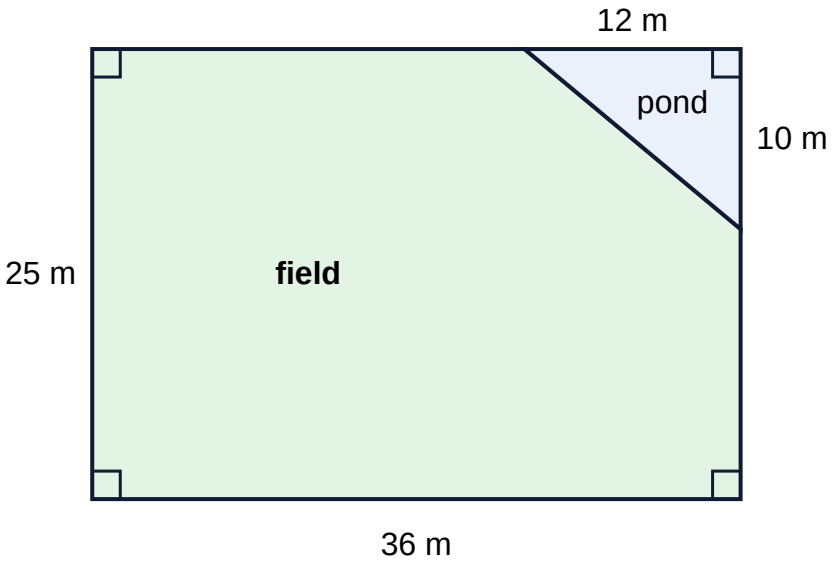
Working

Answer ____ cows

(b) Which ratio is equivalent to 2 : 7? **[1]**

- ☐ A) 4 : 9
- ☐ B) 7 : 2
- ☐ C) 6 : 20
- ☐ D) 8 : 28

8 A field is a rectangle 36 m long and 25 m wide. One corner of it is a pond in the shape of a right-angled triangle. The diagram is not drawn to scale. **[1]**



(a) Work out the area of the pond.

Working

Answer ____ m^2

(b) Work out the area of the field that is **not** pond. Show your working. **[2]**

Working

Answer ____ m^2

9 At the gala, Ali's time for one race is 1.6 minutes. **[1]**

Complete: 1.6 minutes = 1 minute and _____ seconds

Working

Answer ____ seconds

10 Eight swimmers swim 50 m backstroke. Their times, in seconds, are: **[1]**
32.4, 30.9, 33.5, 31.2, 30.9, 34.0, 31.8, 32.1

(a) Find the range of the times.

Working

Answer ____ seconds

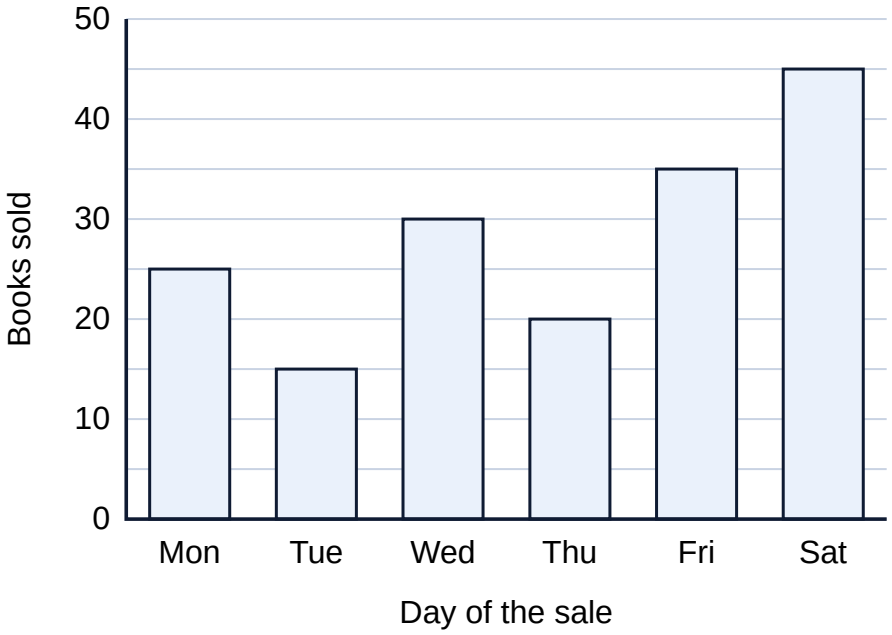
(b) Calculate the mean time. Show your working. **[2]**

Working

Answer ____ seconds

(c) The coach says, "The mode is the best average to describe these times." **[1]**
Do you agree? Give a reason.

11 The bar chart shows the number of books a bookshop sold on each day of its sale. **[1]**



(a) How many more books were sold on Saturday than on Tuesday?

Working

Answer _____ books

(b) Before the sale, the manager predicted, "We will sell more books every day than the day before." **[1]**
Was the prediction correct? Use the bar chart to explain.

12 Lina wants to find out if model cars with bigger wheels roll further after going down a ramp. **[1]**

(a) Which of these is continuous data?

- ☐ A) the colour of each car
- ☐ B) the distance each car rolls
- ☐ C) the number of wheels on each car
- ☐ D) the name of each car's maker

(b) Write the **two** sets of data Lina needs to collect for each car to answer her question. **[1]**

13 At the fun fair, there is a "Lucky Duck" game. You pick a duck; some ducks win a prize. **[1]**
On the first morning, 50 people played and 12 won.
By the end of the week, 500 people had played and 140 had won.

(a) Use the results for the whole week to estimate the probability of winning. Give your answer as a fraction or a decimal.

Working

Answer

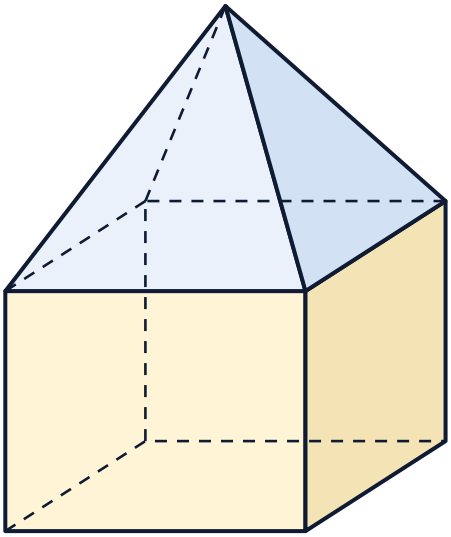
(b) Explain why the results for the whole week give a better estimate than the results for the first morning. **[1]**

(c) Next week, about 2 000 people will play. About how many people would you expect to win? **[1]**

Working

Answer people

14 Amir makes a model garage. The diagram shows its shape. **[1]**



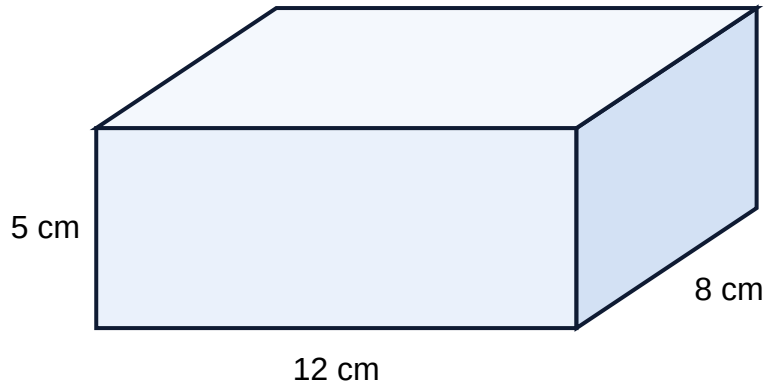
Which two 3D shapes make the model garage?

- ☐ A) a cube and a triangular prism
- ☐ B) a cylinder and a cone
- ☐ C) a cuboid and a square-based pyramid
- ☐ D) a cuboid and a triangular prism

15 **[1]**

Sketch a net of a square-based pyramid. (Draw it on paper.)

16 A model-car kit comes in a closed cardboard box. The box is a cuboid 12 cm long, 8 cm wide and 5 cm high. **[2]**



Work out the total surface area of the box. Show your working.

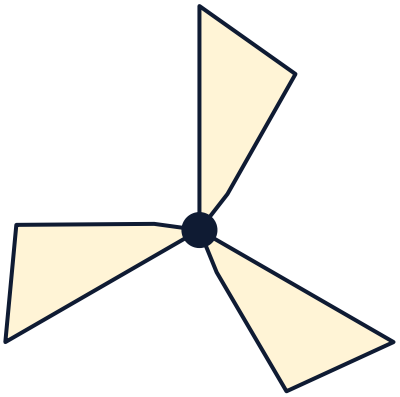
Working

Answer ____ cm^2

17 A bottle of model paint says "capacity 250 ml". **[1]**
Ravi says, "The capacity of the bottle is the same as the volume of the bottle."

Explain the difference between the capacity and the volume of the bottle.

18 This is the logo of the fun-fair windmill ride. **[1]**



What is the order of rotational symmetry of the logo?

Working

Answer

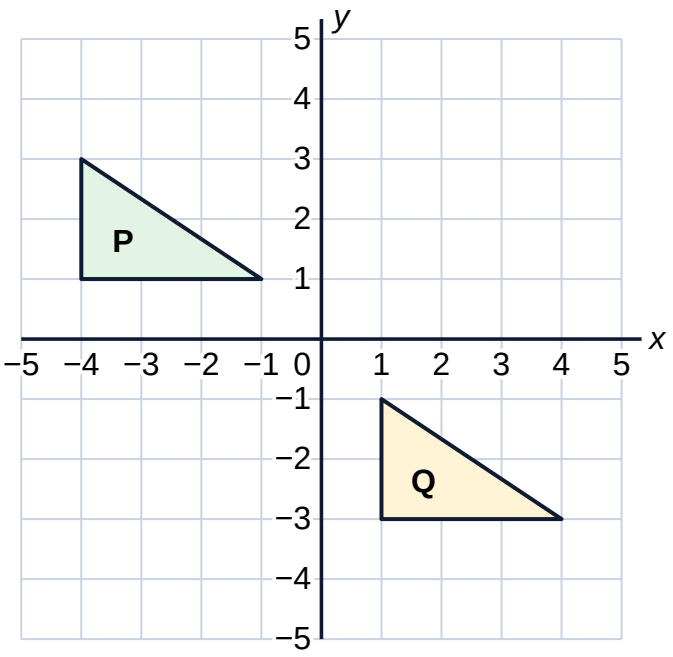
19 **[1]**

Use a protractor to draw an angle of 125°. Write what type of angle it is.

20 A map of the fun fair is drawn on a coordinate grid. **[1]**

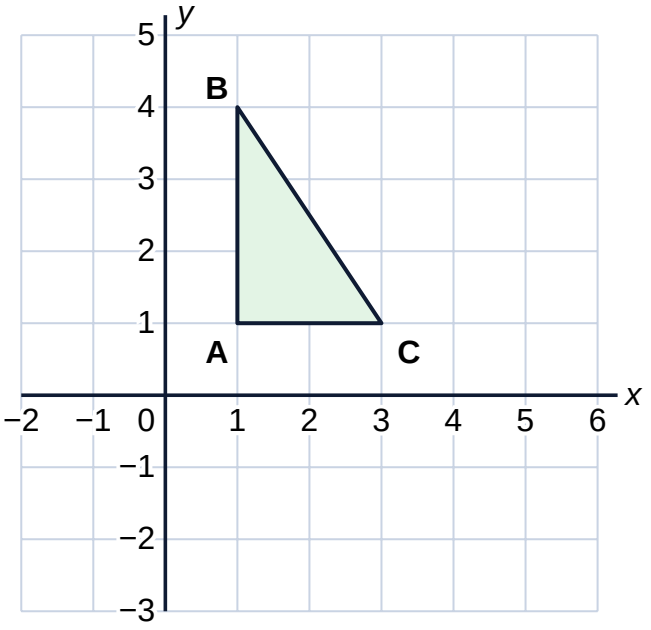
(a) The ice-cream van is at (−2, 3.5). It moves 4 squares to the right and 2 squares down. Write its new coordinates.

[1]



(b) Shape P is a ticket booth. It is moved to position Q.
Describe the translation from P to Q.

21 The diagram shows triangle ABC on a coordinate grid. **[1]**



The triangle is rotated 90° clockwise about the vertex A.
Write the coordinates of the new position of vertex B.

22 On a farm, h is the number of hens and d is the number of ducks. **[1]**
h − d = 12

Write two different pairs of values that h and d could have.

23 The rule for a sequence is: **[1]**
term = position × position + 5

Find the 12th term of the sequence.

Working

Answer

24 Four friends go to the bookshop sale. Together they buy 3 books at $12.80 each and 2 maps at $7.40 each. They share the total cost equally. **[2]**

How much does each friend pay? Show your working.

Working

Answer $

25 A farm sells eggs in boxes of 12. One week the hens lay 950 eggs. **[1]**

How many boxes can be filled completely?

Working

Answer boxes